

is chosen from the agents on island 1. We consider the virality of the article when agents on island 1 share with probability $1 - H(b_1^{**})$ under the stronger homophily structure (p'_s, p'_d) versus (p_s, p_d) (and all other agents kill the article). This is sufficient to show that virality increases following the increase in homophily, because obviously virality with $b_1^{**'} < b_1^{**}$ (but the same network $\mathbf{P}$) is strictly higher, given that agents on island 1 share more often ($1 - H(b^{**'}) > 1 - H(b^{**})$).

We consider the diffusion process of an article on the (p'_s, p'_d) network that starts with an agent on island 1. Let us define a *path* of the diffusion process to be a chain $i^* \rightarrow i_1 \rightarrow i_2 \rightarrow \dots \rightarrow i_z$ representing a sequence of agents who receive the article in this process, with i^* being the seed agent, i_1 through i_{z-1} all being agents who shared it, and agent i_z being an agent who either ignored or disliked the article. There may be many such paths for the diffusion of the article (by assumption, all agents with the possible exception of agent i_z must be on island 1).

For each path, we define an alternative path (generated randomly) as follows. For any links to agents other than to agent i_z (i.e., links within island 1), with probability $(p'_s - p_s)/p'_s$, the link instead goes to one of islands $2, \dots, k$ (chosen in proportion to their population) and otherwise remains the same. Applying this to all paths, we define an isomorphic diffusion process to one on a sharing network with weaker homophily parameters (p_s, p_d) . However, note that the length of every path cannot increase following this transformation. Because any transition to islands $2, \dots, k$ is necessarily the end of the path, paths can only shorten. Moreover, the number of paths must weakly decrease. As a result, the fraction of agents who receive the article (i.e., S_{i^*}) must be lower, and virality is less under the (p_s, p_d) sharing network. This establishes part (a).

For part (b), we first note that there exists $\bar{r}$ such that the most-sharing equilibrium when $r > \bar{r}$ is all-share ($b_\ell^{**} = 0$ for all islands ℓ) regardless of $\mathbf{P}$. Notice that equation (2) is minimized when $b_i = 0$, and in particular, for all agents i (regardless of their prior) $\pi_i \geq \frac{(1-p)\phi(r)}{(1-q)(1-\phi(r)) + (1-p)\phi(r)}$. Then, letting $\bar{\pi} = \max \left\{ \frac{c}{u+c}, \frac{\bar{u}-\bar{c}}{\bar{u}} \right\} < 1$, we note that whenever $r \geq \phi^{-1} \left(\frac{(1-q)\bar{\pi}}{(p-q)\bar{\pi} + (1-p)} \right) \equiv \bar{r} \in (0, 1)$, $\pi_i \geq \bar{\pi}$. Of course, when all other agents (other than i) share and $r > \bar{r}$, $U_i(\mathcal{S}) \geq u\pi_i - c(1 - \pi_i) \geq 0$ and $U_i(\mathcal{D}) = \bar{u}(1 - \pi_i) - \bar{c} \leq 0$, so $a_i = \mathcal{S}$ is a best response for agent i . Thus, the most-sharing equilibrium is all-share (because it is an equilibrium and no other strategy profile can have more sharing).

Observe that when $r > \bar{r}$, virality is measured simply by the expected size of the connected component (formed by $\mathbf{P}$) containing the seed agent i^* . Regardless of the homophily parameters, the seed agent i^* will be chosen from the largest island (call this island ℓ^*). This is immediate from the fact that all agents share in equilibrium, agents on island ℓ^* have the most connections to any other arbitrary island ℓ' (in expectation), and are connected to all agents on their own island.

Lastly, we note that the probability that island ℓ has any connections to island ℓ' is given by $\tilde{p}_{\ell, \ell'} = 1 - (1 - p_d)^{N_\ell N_{\ell'}}$ *before* the decrease in homophily and $\tilde{p}'_{\ell, \ell'} = 1 - (1 - p'_d)^{N_\ell N_{\ell'}}$ *after* the decrease in homophily, with $\tilde{p}'_{\ell, \ell'} > \tilde{p}_{\ell, \ell'}$ for all pairs of islands (ℓ, ℓ') because $p'_d > p_d$. Using the same terminology as in the argument for part (a), we map the diffusion paths of an article under the less homophilic sharing network with (p'_s, p'_d) . Consider cycles between islands $\ell^* \rightarrow \ell_1 \rightarrow \ell_2 \dots \rightarrow \ell_z$, where ℓ_z is the same island as one of $\ell^*, \ell_1, \dots, \ell_z$ (in which case, no additional engagement is obtained thereafter the